Multi-Agent Reinforcement Learning in the Clean Up Environment: Generational Evolution and Exploiter Training

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Abstract

This paper presents a multi-agent reinforcement learning (MARL) study conducted in the Clean Up substrate from the Melting Pot 2.0 benchmark which is a social dilemma environment where agents must balance individual apple harvesting with the collective responsibility of cleaning a polluted river. We investigate and compare two distinct training pipelines for producing agents capable of exploiting cooperative background partners.

The first pipeline follows a generational evolutionary approach: seven agents are trained across ten generations of 1000 episodes each (10000 episodes in total) using Proximal Policy Optimization (PPO) in the base Clean Up substrate, with intergenerational fitness-based selection and weight perturbation driving population-level improvement. The top three agents from the final generation are then transferred into a structured scenario containing four scripted reflex cleaners and fine-tuned for a further 3000 episodes, resulting in a total training budget of 13000 episodes.

The second pipeline is a pure scenario approach: three focal agents are trained from scratch directly in the same scenario for 3000 episodes, after which the best-performing agent is identified, its weights are copied to the remaining two agents with added Gaussian noise, and all three continue training for a further 3000 episodes, for a total of 6000 episodes.

Both pipelines employ extensive reward shaping, action history conditioning, and anti-degenerate penalties to prevent behavioral collapse. Notably, despite requiring less than half the total training experience, the pure scenario pipeline produces agents that outperform those trained via the generational-then-transfer approach, suggesting that direct specialization in the target scenario is more effective than broad pre-training followed by transfer in this setting.

1 Introduction

1.1 Problem and Motivation

Cooperation and free-riding are central tensions in social dilemmas. In many real-world and simulated settings, rational self-interested agents may forgo contributing to a public good in favor of exploiting the contributions of others. Understanding when and how agents learn such strategies, and how the training regime shapes the resulting policies, is a core challenge in multi-agent reinforcement learning. The Clean Up environment from Melting Pot 2.0 [1] instantiates this dilemma concretely: a river becomes polluted over time, reducing the rate at which apples spawn on an adjacent field. Agents can choose to clean the river (a collective contribution yielding no direct individual reward) or harvest apples (an individual payoff). Because cleaning is individually costly but collectively beneficial, a purely self-interested agent is incentivized to free-ride on the cleaning efforts of others.

1.2 Objective

This project aims to determine whether pre-training agents in the full multi-agent substrate via genrational evolution, followed by scenario fine-tuning, yield better exploiters than training focal agents directly in the target scenario. Moreover, we seek to investigate what reward and training design choices are necessary to prevent degenerate behaviors and encourage meaningful exploitation of cooperative partners.

2 Related Work

2.1 Melting Pot 2.0 (Agapiou et al.. 2023)

Agapiou et al. [1] introduce Melting Pot 2.0, a suite of multi-agent evaluation scenarios built on DeepMind Lab2D that stresses generalization to novel social contexts by separating training from evaluation partners. The Clean Up substrate used in this work is part of this benchmark. Our use of the scenario API with scripted background agents mirrors the Melting Pot evaluation philosophy, though we repurpose it for adversarial fine-tuning. The background agents follow a fixed cycle — cleaning for 200 steps then eating for 200 steps — providing a controlled and predictable cooperative partner whose behavior the focal agents are tasked with exploiting.

2.2 PPO in Cooperative Multi-Agent Games (Yu et al., 2022)

Yu et al. [2] demonstrate that independent PPO — where each agent runs its own update with no explicit coordination mechanism — achieves surprisingly strong performance in cooperative multi-agent tasks, often matching or outperforming centralized training algorithms. This motivates our use of independent PPO per agent throughout all training phases.

3 Agent Description, Methods, and Experimental Setup

3.1 Environment

All experiments are conducted in the Clean Up substrate from Melting Pot 2.0. The base substrate (accessed via the shimmy compatibility wrapper) is used in the generational phase, while the clean_up_2 scenario is used in both scenario training pipelines.

Each agent receives a RGB observation of size 88×88×3 and selects from a discrete action space of 9 actions: NOOP (0), FORWARD (1), BACKWARD (2), LEFT (3), RIGHT (4), TURN_LEFT (5), TURN_RIGHT (6), ZAP (7), and CLEAN (8).

The apple spawn rate is inversely proportional to river pollution. Eating an apple yields a raw environment reward of +1.0. Cleaning yields no direct reward. The background agents in the scenario alternate between cleaning for 200 steps and eating for 200 steps per episode.

3.2 Network Architecture

All agents across both pipelines share the same AgentNetwork architecture.

CNN Encoder. Three convolutional layers (32, 64, 64 filters; kernels 8×8, 4×4, 3×3; strides 4, 2, 1), each followed by ReLU, then flattened into a compact visual feature vector.

Action History Conditioning. The most recent actions of all agents in the population are one-hot encoded and concatenated with the CNN features, allowing each agent to condition its policy on peer behavior. The dimension varies by phase: 7×9=63 in the generational phase, 3×9=27 in the scenario phase. In the generation-to-scenario transfer (Pipeline A), N_AGENTS_TRAIN=7 is retained for checkpoint compatibility, padding unobserved background agent slots with zeros.

Policy Backbone. Two fully connected layers (256 hidden units, ReLU).

Actor Head. Linear → 9 action logits → Categorical distribution.

Critic Head. Linear → scalar state value for PPO.

3.3 Training Algorithm

All agents use independent PPO [2] with Generalized Advantage Estimation (GAE). Each agent maintains its own policy, value function, optimizer, and gradient scaler. Mixed precision training (torch.amp) is used throughout for GPU efficiency.

3.4 Reward Shaping

Apple reward amplification. Raw rewards ≥ 1.0 are scaled to $+25.0$.
River proximity detection. A pixel-wise water mask in a fixed ROI of each agent's observation determines an ROI score: $+1$ (at river), 0 (nearby), -1 (far). The mask fires when $(G + B) > 1.5R$.
Cleaning rewards. Attempting CLEAN at the river: $+2.0$. Successful clean (local clean fraction increases $> 10\%$): additional $+15.0$. Non-NOOP action near river: $+0.05$. Movement (FORWARD/LEFT/RIGHT) near river: additional $+0.2$.
Anti-degenerate penalties. ZAP: -2.0. NOOP: -1.0.
Action queue penalties (rolling buffer of 50 actions). Spinning (alternating turns with no other actions): -2.0. No CLEAN in past 20 steps: -0.5.

3.5 Hyperparameters

Hyperparameter	Pipeline A – Generational	Pipeline A – Scenario	Pipeline B – Phase 1	Pipeline B – Phase 2
Number of focal agents	7	3	3	3
Episodes	10,000	3,000	3,000	3,000
Steps per episode	700	700	700	700
Training epochs per rollout	6	6	6	6
Minibatch size	1,024	512	512	512
Discount factor γ	0.99	0.99	0.99	0.99
GAE λ	0.97	0.95	0.95	0.95
PPO clip ratio ϵ	0.2	0.2	0.2	0.2
Value coefficient	0.5	0.5	0.5	0.5
Gradient clip norm	0.5	0.5	0.5	0.5
Learning rate	1×10^{-3}	1×10^{-3}	1×10^{-3}	1×10^{-3}
Entropy schedule	$0.7 \to 0.3$	$0.3 \to 0.15$	$0.7 \to 0.3$	$0.3 \to 0.15$
FC hidden units	256	256	256	256

Table 1: Training hyperparameters for both pipelines.

4 Training Pipelines

4.1 Pipeline A - Generational Evolution + Scenario Fine-Tuning

4.1.1 Phase A1 - Generational Training

Seven agents train concurrently in the full Clean Up substrate for 10 generations of 1,000 episodes each (10,000 total episodes). Entropy is annealed from 0.7 to 0.3 across the full generational phase. After each generation, a FitnessTracker computes a composite score per agent using normalized per-step rates:

$$\begin{aligned} \text{fitness} &= 3.0 \times \text{apple\_rate} + 2.0 \times \text{good\_clean\_rate} \\ &\quad + 0.5 \times \text{clean\_attempt\_rate} + 0.3 \times \text{river\_proximity\_rate} \\ &\quad - 2.0 \times \text{zap\_rate} - 0.5 \times \text{noop\_rate}. \end{aligned} \tag{1}$$

The population is evolved after each generation:

- **Rank 1 & 2 (elites):** weights unchanged.

- **Rank 3 & 4:** copied from rank 1 plus small Gaussian noise ($\sigma = 0.01$).
- **Rank 5, 6 & 7:** copied from rank 1 plus large Gaussian noise ($\sigma = 0.05$).

4.1.2 Phase A2 - Scenario Fine-Tuning

The three highest-fitness agents from generation 9 are loaded into clean_up_2 scenario and trained alongside the scripted background cleaners for 3000 episodes only. Entropy is annealed 0.3 to 0.15 across these 3000 episodes.

4.2 Pipeline B: Pure Scenario Training

4.2.1 Phase B1 - From-Scratch Scenario Training

Three focal agents are initialized from random weights and trained directly in the clean_up_2 scenario for 3000 episodes. Entropy is annealed from 0.7 to 0.3 across this phase.

4.2.2 Phase B2 - Anchor-Copy Continuation

After 3000 episodes, the best-performing focal agent is identified. Its weights are copied in the remaining two agents with added Gaussian noise, and all three continue training for a further 3000 episodes. Entropy is annealed from 0.3 to 0.15 across this second phase.

4.3 Pipeline Comparison

Aspect	**Pipeline A**	**Pipeline B**
Initial training environment	Full substrate (7-agent self-play)	Scenario (3 focal + 4 scripted)
Pre-training before scenario	10000 episodes	None
Scenario training episodes	3000	6000
Total training episodes	13000	6000
Anchor-copy mechanism	No	Yes (after 3000 eps)
Entropy schedule	$0.7 \rightarrow 0.3 \rightarrow 0.15$	$0.7 \rightarrow 0.3 \rightarrow 0.15$

Table 2: Comparison of the two training pipelines.

5 Results

5.1 Pipeline A

5.1.1 Cross-Environment Evaluation of Generationally Trained Agents

We evaluate the final-generation agents produced by the generational evolutionary pipeline across two environments:

1. The base `clean_up` substrate (7-agent self-play).

2. Scenario 2 (`clean_up_2`), containing 3 focal agents.

All agents were taken from generation 10. Each setting was tested over 10 episodes of 700 steps.

Evaluation in the Clean Substrate

Seven agents were evaluated together in the original full environment. Results are shown in Table 3.

Ep	A0	A1	A2	A3	A4	A5	A6	Mean
0	0	0	0	0	0	0	0	0.0
1	0	0	0	0	0	0	0	0.0
2	0	0	0	0	0	0	0	0.0
3	0	0	0	0	0	0	0	0.0
4	0	0	0	0	0	0	0	0.0
5	0	0	0	0	0	0	0	0.0
6	0	0	0	0	0	0	0	0.0
7	0	19	0	17	0	1	0	5.29
8	0	0	0	0	0	0	0	0.0
9	0	0	0	0	0	0	0	0.0
Mean	0.0	1.9	0.0	1.7	0.0	0.1	0.0	0.53

Table 3: Evaluation of generation-trained agents in the clean substrate.

Performance is extremely sparse. Across 10 episodes, nearly all returns are zero. The overall mean return is 0.53, indicating that generational training did not produce robust harvesting behavior under unrestricted seven-agent self-play.

Evaluation in Scenario 2

We next evaluated three selected generation-10 agents (IDs 1, 2, 6) in Scenario 2. Results are shown in Table 4.

Ep	A0	A1	A2	Mean
0	7	53	22	27.33
1	18	59	2	26.33
2	101	114	60	91.67
3	56	72	3	43.67
4	9	0	26	11.67
5	4	137	0	47.00
6	60	25	44	43.00
7	53	57	46	52.00
8	7	58	20	28.33
9	1	19	64	28.00
Mean	31.6	59.4	28.7	39.9

Table 4: Evaluation of generation-trained agents in Scenario 2.

In stark contrast to the clean substrate results, agents achieve a mean return of 39.9. Returns are consistently positive, with peak episode performance exceeding 90 mean return.

Cross-Environment Comparison

The performance gap is substantial:

$$\text{Mean return (clean)} = 0.53$$

$$\text{Mean return (scenario)} = 39.9$$

This corresponds to an approximately $75\times$ increase in average reward when the same policies are evaluated in the structured scenario.

Interpretation. The generational evolutionary pipeline appears unable to produce stable harvesting behavior under full seven-agent competition. However, when placed in the more structured and lower-interference Scenario 2, the same policies demonstrate strong reward acquisition.

5.1.2 Evaluation After Additional Scenario Training

After completing 10 generations of evolutionary training in the clean substrate, the three highest-fitness agents were selected and further trained for 3,000 episodes directly in Scenario 2 (`clean_up_2`). These agents were then evaluated in the same scenario for 10 episodes using stochastic action sampling.

Table 5 reports the per-agent mean returns.

Agent	A0	A1	A2	Overall Mean
Mean Return	29.4	57.1	36.8	41.1

Table 5: Performance of generational agents after 3,000 additional Scenario 2 training episodes.

The agents achieve an overall mean return of 41.1, indicating strong harvesting performance within the structured scenario environment. Compared to zero-shot transfer from generational training alone, performance increases modestly but becomes more consistent across episodes.

Notably, agent specialization emerges more clearly after fine-tuning, with Agent A1 achieving the highest mean return (57.1). This suggests that additional scenario training refines and stabilizes the policies learned during evolutionary self-play.

5.2 Pipeline B

5.2.1 Evaluation in Scenario 2

Pipeline B agents were trained directly in Scenario 2 (`clean_up_2`), consisting of three focal learning agents and four scripted background agents alternating between cleaning and harvesting phases. No pre-training in the full substrate was performed.

After training, the three focal agents were evaluated for 10 episodes. Table 6 reports the per-agent and overall mean returns.

Episode	A0	A1	A2	Mean
0	22.0	16.0	125.0	54.3
1	22.0	0.0	149.0	57.0
2	29.0	57.0	101.0	62.3
3	7.0	33.0	12.0	17.3
4	13.0	44.0	26.0	27.7
5	54.0	60.0	13.0	42.3
6	0.0	48.0	16.0	21.3
7	0.0	0.0	46.0	15.3
8	59.0	106.0	126.0	97.0
9	0.0	30.0	46.0	25.3
Mean	**20.6**	**39.4**	**66.0**	**42.0**

Table 6: Evaluation results for Pipeline B (scenario-only training) in Scenario 2 over 10 episodes.

The overall mean return across all agents and episodes is 42.0. Agent A2 achieves the highest individual mean return (66.0), indicating strong specialization toward harvesting under the structured background-agent dynamics.

However, performance variance across episodes is substantial, with several low-return episodes (e.g., Episodes 3 and 7) suggesting sensitivity to early coordination dynamics or stochastic policy effects. Compared to generationally pre-trained agents, the scenario-only pipeline achieves comparable overall performance but exhibits stronger inter-agent asymmetry, with one dominant agent contributing disproportionately to total returns.

6 Conclusion

This work compared two training pipelines for multi-agent learning in Scenario 2 of the Clean Up environment: (i) a generational self-play pipeline followed by scenario fine-tuning (Pipeline A), and (ii) direct scenario-only training (Pipeline B).

The primary finding is that both pipelines achieve highly comparable final performance. The overall mean return after evaluation is approximately 41.1 for Pipeline A (after generational pre-training and additional scenario training) and 42.0 for Pipeline B (trained directly in the scenario). Despite their fundamentally different training procedures, both approaches converge to similar reward levels in the target environment.

Crucially, Pipeline B reaches this performance using substantially fewer total training episodes. While Pipeline A requires 13,000 episodes in total (10000 generational + 3000 scenario), Pipeline B achieves comparable results with only

6000 scenario episodes. This suggests that, when the deployment environment is known in advance, direct scenario training can be significantly more sample-efficient than first training in a broader substrate and transferring.

However, qualitative differences emerge in agent behavior. Pipeline B exhibits stronger inter-agent asymmetry, with one agent contributing disproportionately to the total return. In contrast, generational pre-training appears to encourage more balanced performance across agents, likely due to exposure to more diverse self-play dynamics during evolution. This indicates that although total reward may converge to similar levels, the structure of cooperation and specialization can differ meaningfully between pipelines.

Overall, these results suggest that scenario-specific training is a highly efficient strategy when the target environment is fixed, while generational pre-training may be advantageous when robustness, behavioral diversity, or transferability to new scenarios is desired. Future work could explore hybrid approaches that combine the sample efficiency of scenario training with the behavioral robustness induced by generational self-play.

References

[1] Agapiou, John P., et al. "Melting Pot 2.0." arXiv.Org, 31 Oct. 2023, `https://arxiv.org/abs/2211.13746`.

[2] Yu, Chao, et al. "The Surprising Effectiveness of PPO in Cooperative, Multi-Agent Games." arXiv.Org, 4 Nov. 2022, `https://arxiv.org/abs/2103.01955`.